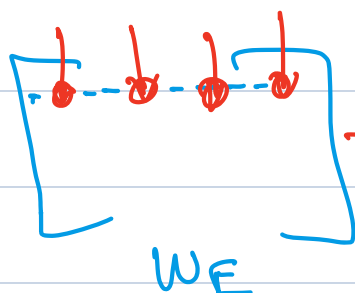


Core Components of a Transformer.

$$\begin{bmatrix} W_E \end{bmatrix}_{n-d \times n-v}$$

$n-d$ - hidden value
- Vect/Token dim.

$\begin{matrix} T_1 & T_2 & T_3 & T_4 \\ | \text{The} | & | \text{Cat} | & | \text{sat} | & | \text{on} | \\ \text{Seq} \end{matrix} \rightarrow \text{map}$



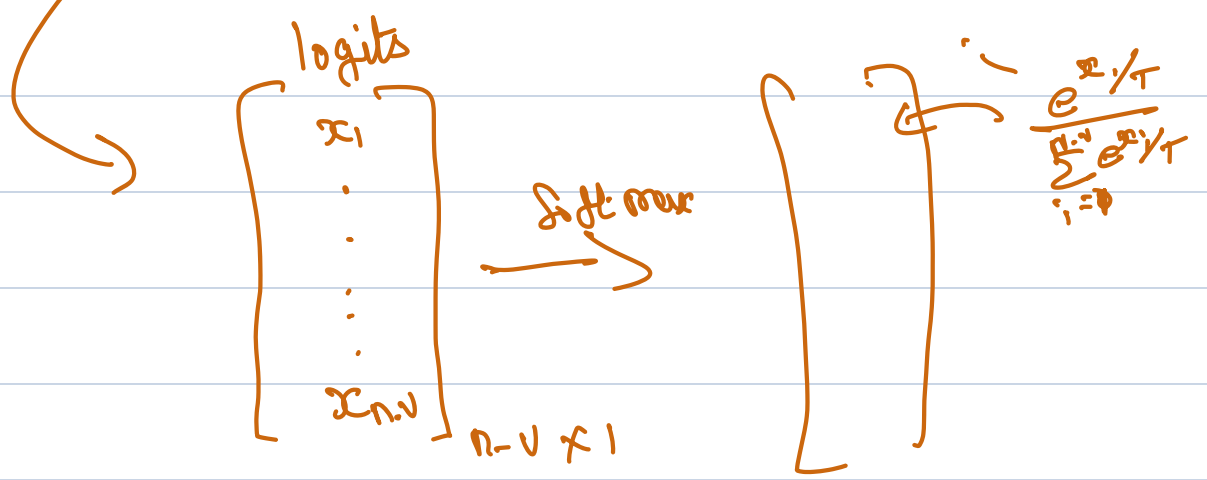
$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n-d \times s-d} \quad W_E \text{ (only with Seq top)}$$

$s-d \leq \text{Context length of the model!}$
 → [How many tokens map it can attend]



unembedding matrix W_O

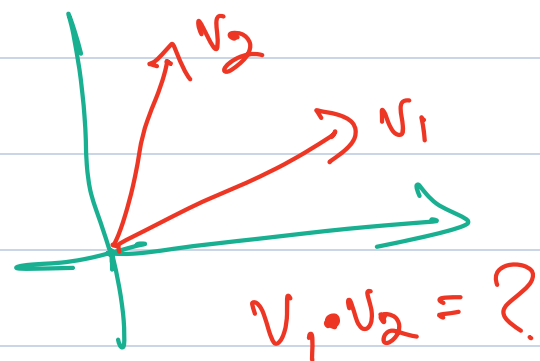
$$\begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n-v \times n-d} \rightarrow \begin{bmatrix} \vdots \\ \vdots \\ \vdots \\ \vdots \end{bmatrix}_{n-d \times 1}$$



Attention

$$\begin{bmatrix} w_q \end{bmatrix}_{d_q \times n_d} \begin{bmatrix} T_i \end{bmatrix}_{n_d \times 1} \rightarrow \begin{bmatrix} a_i \end{bmatrix}_{d_q \times 1}$$

$$\begin{bmatrix} w_k \end{bmatrix}_{d_k \times n_d} \begin{bmatrix} T_i \end{bmatrix}_{n_d \times 1} \rightarrow \begin{bmatrix} k_i \end{bmatrix}_{d_k \times 1}$$

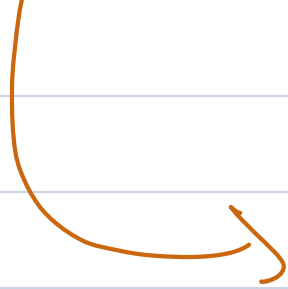


$$\text{Soft max} \left(\frac{k \cdot Q}{\sqrt{d-k}} \right) = \begin{bmatrix} \\ \\ \\ \end{bmatrix}_{4 \times 4}$$

each value $\in (0,1)$ in a column.

Next

$$\begin{bmatrix} W_v \\ \text{Low rank rep.} \\ [n_d \times n_v] \end{bmatrix} \begin{bmatrix} T_i \\ \text{rank } n_d, \\ n_v \times n_d \end{bmatrix} \begin{bmatrix} T_i \\ \text{rank } n_d, \\ n_d \times n_d \end{bmatrix}$$



$$\begin{bmatrix} \vdots \\ V_i \end{bmatrix}$$

$n-d+1$

$$T_{ij} = \sum_{i=1}^{n-d} U_i$$